

ModemBridge-2k26 User Guide

ModemBridge-2k26 is a modern, fleet-aware serial-to-network bridge. It emulates a Hayes-compatible dial-up modem, allowing retro computers (like the Atari 8-bit series) to connect to modern TCP Telnet and SSH BBS systems over standard serial connections.

Operating entirely from your System Tray, it allows you to manage multiple serial ports simultaneously, host inbound BBS callers, and monitor real-time traffic via a local Web Dashboard.

1. The System Tray Interface

ModemBridge runs quietly in the background. Left-click or Right-click the app icon in your system tray to access the **Fleet Menu**.

- **Active Bridges:** Each enabled serial port appears as its own sub-menu (e.g., [Atari 800 Node 1]).
 - **Quick Actions:** Inside each bridge's sub-menu, you can quickly open the visual Phonebook to dial out, send your User ID/Password macros, or force a Hangup.
 - **Global Controls:** Access the Global Settings or launch the Web Dashboard.
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2. Fleet Configuration (Global Settings)

Click **Global Settings...** in the tray menu to configure your serial ports and web servers. ModemBridge features **Surgical Hot-Swapping**—meaning you can tweak settings, change inbound ports, or load new phonebooks, and clicking "Save" will apply them *instantly* without dropping your active retro connections!

Adding and Configuring Ports

1. **Select a Port:** Choose an OS-detected port (e.g., COM1 or /dev/ttyUSB0) from the left pane, or click **Add Custom...** for virtual ports.

2. **Enable Bridge:** Check "Enable this Bridge" to bring it online.
3. **Friendly Name:** Assign a human-readable name (e.g., "Main Atari Console").
4. **Hardware Settings:** Select your baud rate (1200 to 115200), and toggle RTS/CTS Hardware Flow Control and Local Echo based on your terminal software's needs.
5. **Phonebook XML:** Browse for or create a new XML file to store your favorite BBS addresses and login credentials.

Web Server Ports

At the bottom of the Settings dialog, you can assign the **HTTP Web Port** (default 8080) and the **WebSocket Data Port** (default 12345). Ensure these do not conflict with other services running on your PC.

3. Dialing Out (Connecting to a BBS)

Once your terminal software is running on the retro computer, you can dial out using standard Hayes AT commands.

- **Standard Telnet:** Type ATDT hostname.com:23
 - **Secure SSH:** Type ATDT ssh:username@hostname.com:22. If the SSH server requires a password, the bridge will pause and prompt you with PASSWORD: in your terminal.
 - **Dialing from the Phonebook:** If you have an entry in your XML phonebook named "Level29", simply type ATDT Level29. The bridge will handle the IP, port, protocol, and credentials automatically.
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4. Hosting an Inbound BBS (TCP Listener)

ModemBridge can act as a server, allowing external users to Telnet into your retro machine.

Setup & Disabling the Listener

In the Global Settings, you can assign an **Inbound BBS Port** (e.g., 2323) to your chosen serial port.

- **To Enable:** Enter any valid, unused port number between 1 and 65535.
- **To Disable:** Set the port to **0**. This instantly shuts down the inbound TCP listener for this specific bridge, while keeping your outbound serial connection and terminal perfectly intact.

Answering a Call

1. When a user connects to your assigned port via Telnet, ModemBridge will send RING to your terminal screen every 3 seconds.
2. **Manual Answer:** Type ATA to pick up the line. The bridge will output CONNECT and hand off the data stream.
3. **Auto-Answer:** Type ATS0=2 (or any number of rings) to tell the modem to answer automatically.

Note: ModemBridge features **Smart Call Rejection**. If your Atari is currently in a call, dialing out, or already ringing, any new inbound connections are immediately dropped with a BUSY signal to protect your active session.

5. In-Call Features & Macros

While actively connected to a remote system, ModemBridge intercepts specific keystrokes for quick actions:

- **Escape to Command Mode:** Pause for 1 second, type +++, and pause again. The bridge will respond OK, allowing you to send AT commands without dropping the connection. Type ATZ to drop the line.
- **Macro Keys:** If you press the physical Escape key on your Atari, the bridge enters a "listening" state for your next keystroke:

- Esc then U: Automatically injects the Username associated with your current Phonebook connection.
 - Esc then P: Automatically injects the Password.
 - Esc then H: Forces an immediate hardware hangup.
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6. The Web UI Dashboard

Select **View Dashboard** from the tray menu to open the Web UI in your default browser.

The dashboard provides a real-time, multi-node command center:

- **Live Traffic:** Watch color-coded TX (Transmit) and RX (Receive) data flows for every active bridge.
 - **State Monitoring:** See exactly which nodes are OFFLINE, READY, DIALING, CONNECTED, or RINGING.
 - **Remote Control:** Dial a BBS, send credentials, or drop calls for any node directly from your modern PC browser, without touching the retro hardware.
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7. Hayes AT Command Quick Reference

Command	Description	Example
AT	Attention (Ping the modem)	AT (Returns OK)
ATDT	Dial Target	ATDT particlesbbs.dyndns.org:6400
ATDT	Dial SSH Target	ATDT ssh:pi@192.168.1.50:22
ATA	Answer Inbound Call	ATA
ATS0=x	Set Auto-Answer Rings	ATS0=3 (Answers after 3 rings)
ATH	Hangup immediately	ATH

Command	Description	Example
ATZ	Reset modem & drop carrier	ATZ